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Personalized Student Behavior Profiling Through Machine Learning Clustering

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ABSTRACT

This study presents an application of machine learning techniques to analyze and cluster the learning behaviors of students at the Vientiane-Hanoi Friendship Technical Vocational College. Utilizing a dataset comprising 756 students and six academic performance metrics (Class Participation, Rules Adherence, Activities, Tests, Homework, and Exams), this research compares the efficacy of three distinct clustering algorithms: K-Means, Agglomerative Clustering, and DBSCAN. The data underwent preprocessing, including feature scaling via Standardization and dimensionality reduction using Principal Component Analysis (PCA) for visualization. The analysis successfully identified three primary student archetypes: (1) a high-achieving group, (2) an average or mixed- performance group, and (3) a high-risk, low-performing group. Model performance, evaluated using the Silhouette Score, revealed that DBSCAN (score: 0.7301) significantly outperformed K-Means (0.3586) and Agglomerative Clustering (0.2933), primarily due to its ability to effectively identify and isolate outliers. The findings provide actionable insights for educational institutions to develop targeted intervention strategies, thereby enhancing student support and improving overall academic outcomes.

Keywords: Educational Data Mining, Student Clustering, K-Means, DBSCAN, Agglomerative Clustering, Principal Component Analysis

1. Introduction

In the modern educational landscape, understanding student performance extends beyond final grades [1]. The ability to identify underlying behavioral patterns is crucial for providing timely and effective academic support. For institutions like the Vientiane-Hanoi Friendship Technical Vocational College [2], recognizing at-risk students or distinct learning profiles within a large student body can enable targeted interventions that significantly improve educational success rates. Traditional analysis methods often fail to uncover the complex, nonlinear relationships hidden within educational data.

Machine learning, particularly unsupervised clustering techniques [3], offers a powerful solution for automatically identifying natural groupings within data without predefined labels. By analyzing various facets of student performance from classroom engagement to examination results clustering algorithms can reveal distinct archetypes of learners [4].

This study aims to apply and compare three prominent clustering algorithms K-Means [5], Agglomerative Clustering [6], and DBSCAN [7] to a dataset of student performance from the Vientiane-Hanoi Friendship Technical Vocational College. The primary objectives are: (1) to determine the optimal number of distinct student clusters within the dataset, (2) to characterize the academic profile of each identified cluster, and (3) to evaluate which algorithm provides the most meaningful and coherent grouping for this specific educational context. The results are intended to equip the institution with a data-driven framework for enhancing pedagogical strategies and student support systems.

2. Methodology

2.1 Dataset Description

The study utilized a dataset from the Vientiane-Hanoi Friendship Technical Vocational College, containing records for 756 students. For each student, six features representing different aspects of academic performance were recorded on a scale of 0-100, first 5 records are shown in Table 1.

Table 1 First 5 records of dataset

Index	Class	Rules	Activities	Test	Homework	Exam
0	40.0	0.0	0.0	0.0	27.5	70.0
1	95.0	100.0	100.0	100.0	100.0	100.0
2	85.0	90.0	90.0	90..0	90.0	90.0
3	90.0	100.0	95.0	90.0	100.0	90.0

4	95.0	95.0	90.0	0.0	90.0	90.0

Table 1 provides a description of the six features used in this study to evaluate student academic performance.

Class: Score for classroom participation and enthusiasm.

Rules: Score for adherence to classroom and college regulations.

Activities: Score for participation in in-class activities, such as group work and presentations.

Test: Score from quizzes and mid-term assessments.

Homework: Score for timely and complete submission of assignments. Exam: Score from the final semester examination.

2.2. Data Preprocessing

To prepare the data for modeling, two key preprocessing steps were performed.

As clustering algorithms that rely on distance metrics (like K-Means) are sensitive to the scale of features, we applied standardization (Z-score normalization). This process rescales each feature to have a mean of approximately 0 and a standard deviation of 1, ensuring that all features contribute equally to the analysis.

To facilitate visualization of the clustering results in a 2D space, Principal Component Analysis (PCA) [8] was employed. PCA transformed the original 6-dimensional feature space into a 2-dimensional space defined by two principal components (PC1 and PC2) that capture the maximum possible variance in the data.

2.3. Clustering Implementation

The Elbow Method was used to identify the optimal number of clusters (K). This method involves plotting the Within-Cluster Sum of Squares (WCSS) against different values of K [9]. The "elbow" of the plot, where the rate of WCSS decrease sharply slows, suggests the most appropriate value for K in Fig. 1. For our dataset, the elbow point was clearly observed at K=3.

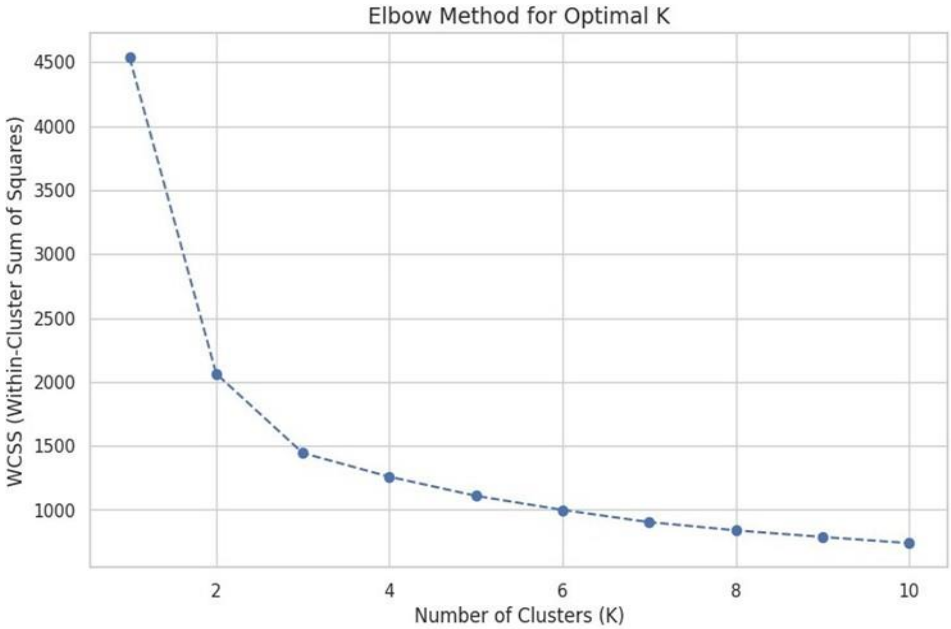

Fig. 1 Elbow method for the optimal number of clusters

Based on the K=3 finding, three clustering models were implemented:

A partitional algorithm, K-Means clustering configured to find 3 clusters (`n_clusters=3`). The k-means++ initialization method was used for more robust centroid selection, with random_state=42` for reproducibility.

A hierarchical algorithm, Agglomerative clustering: set to produce 3 final clusters (`n_clusters=3`). The ward linkage criterion was chosen, which aims to minimize the variance within the clusters being merged.

This algorithm, DBSCAN (Density-Based Spatial Clustering of Applications with Noise) does not require a predefined number of clusters. It was configured with eps=1.2 (the maximum distance between two samples for one to be considered as in the neighborhood of the other) and min_samples=12 (the number of samples in a neighborhood for a point to be considered a core point). A key feature of DBSCAN is its ability to identify noise points (outliers), which are labeled as cluster -1.

2.4. Evaluation Metric

The performance of each clustering model was quantitatively assessed using the Silhouette Score [10]. This metric measures how similar an object is to its own cluster compared to other clusters. The score ranges from -1 to +1, where a high value indicates that the object is well-matched to its own cluster and poorly matched to neighboring clusters.

3. Results

3.1. Cluster Profiles by Grade Distribution

Heatmaps were generated to visualize the grade distribution for the clusters identified by each algorithm (Fig. 2).

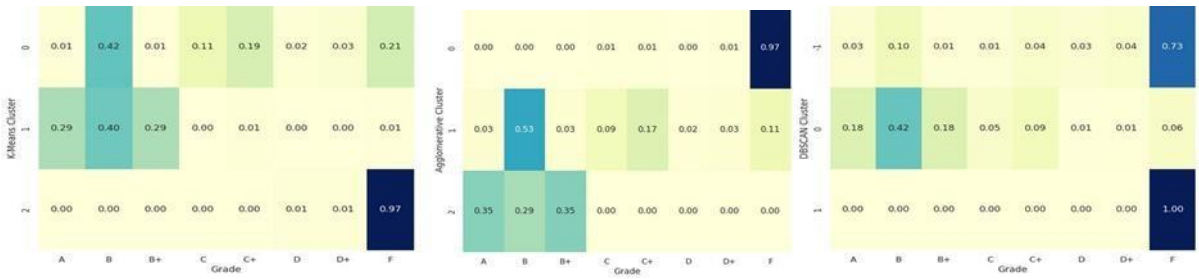

Fig. 2 Grade Distribution Heatmaps per Cluster, (a) K-Means, (b) Agglomerative, (c) DBSCAN

K-Means (Fig. 2a), first cluster 0 (Average/Mixed-Performance) is characterized by a high proportion of 'B' grades (42%) but also significant proportions of 'F' (21%) and 'C+' (19%) grades, indicating a diverse group. Next cluster 1 (High-Achieving) is dominated by students with 'A' (29%), 'B' (40%), and 'B+' (29%) grades. Finally cluster 2 (High-Risk) is an extremely distinct group where 97% of students received an 'F' grade.

Agglomerative Clustering (Fig. 2b), first cluster 0 (High-Risk) is similar to K-Means, this group consisted almost entirely of students with 'F' grades (97%). Second cluster 1 (Average-Performance) is predominantly 'B' grades (53%), with a notable presence of 'C+' (17%) and 'F' (11%) grades. And third one is cluster 2 (High-Achieving) that is a clearly defined group of top performers with grades distributed across 'A' (35%), 'B' (29%), and 'B+' (35%), with no grades below B+.

DBSCAN (Fig. 2c), cluster 1 (High-Risk Core) is a perfectly defined cluster where 100% of students received an 'F' grade. Cluster 0 (High-Achieving Core) is a strong academic group with a majority of 'B' grades (42%) and significant shares of 'A' (18%) and 'B+' (18%). However, cluster -1 (Outliers/Noise) is a heterogeneous group primarily composed of students with 'F' grades (73%) but also containing a scattered mix of all other grades, representing students who did not fit neatly into the two core dense clusters.

3.2. Visual Cluster Separation via PCA

The clustering results were plotted on the 2D space generated by PCA to visualize the separation of groups (Fig. 3). All three models successfully separated a distinct group on the left side of the plot (low PC1 scores, corresponding to low-performing students) from the main body of

students on the right (higher PC1 scores). DBSCAN uniquely identified many points between and around the main clusters as outliers (colored purple).

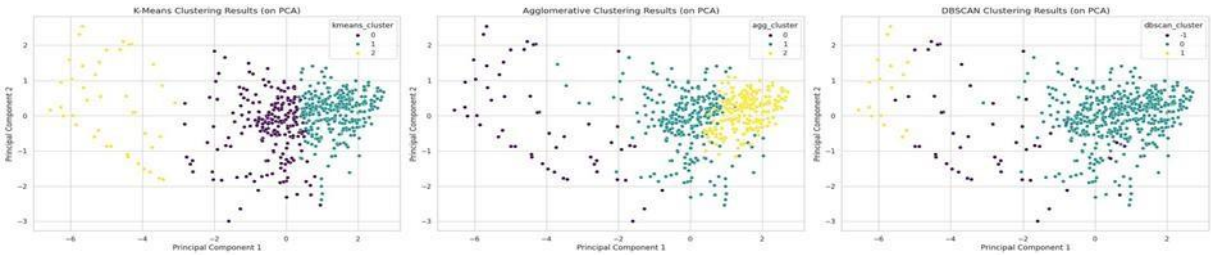

Fig. 3 Clustering Results Visualized on PCA Components, (a) K-Means, (b) Agglomerative, (c) DBSCAN

3.3. Model Performance Evaluation

The Silhouette Scores for each model are presented in Table 2. For DBSCAN, the score was calculated excluding the noise points, as is standard practice.

Table 2 Silhouette Score Comparison

Clustering Model	Silhouette Score
K-Means:	0.3586
Agglomerative	0.2933
DBSCAN (excluding noise)	0.73.1

The results clearly indicate that DBSCAN achieved a significantly higher score, suggesting it produced the most dense and well-separated clusters.

4. Discussion

The analysis confirms that the student population can be meaningfully segmented into three archetypes: a High-Achieving group, an Average/Mixed-Performance group, and a HighRisk group. This finding is consistent across all three algorithms, though the quality and nature of the clusters differ.

The superior performance of DBSCAN (Silhouette Score of 0.7301) is the most significant finding. Its ability to identify and exclude outliers (Cluster -1) is its key advantage. In real-world educational data, students may have unique circumstances that cause their performance profiles to deviate from the norm. While K-Means and Agglomerative Clustering are forced to assign these outliers to one of the predefined clusters (potentially skewing the cluster's

characteristics), DBSCAN correctly flags them as distinct. This makes the resulting core clusters (Cluster 0 and 1) more "pure," cohesive, and representative of the main student populations.

The practical implications for the Vientiane-Hanoi Friendship Technical Vocational College are substantial. (1) The institution can now automatically identify students belonging to the high-risk cluster. This allows for proactive interventions, such as academic counseling, tutoring programs, or meetings with instructors, to address their specific challenges before they fail the course. (2) The average/mixed-performance" group, with its wide grade distribution, may benefit from pedagogical approaches that boost motivation and engagement. Conversely, the high-achieving group can be challenged with advanced projects or enrichment opportunities to foster their potential. (3) The outliers identified by DBSCAN represent unique cases. These students warrant individual investigation to understand the specific reasons for their anomalous performance patterns, which could range from a single missed exam to more complex personal issues.

5. Conclusion

This study successfully demonstrated the application of machine learning clustering techniques to identify meaningful student groups based on academic performance data.

Through a comparative analysis, DBSCAN was identified as the most effective algorithm for this dataset, yielding well-defined clusters and providing the crucial capability of outlier detection. The research provides a robust, data-driven methodology for educational institutions to move beyond one-size-fits-all approaches and develop tailored strategies that support the diverse needs of their student body. By leveraging these insights, institutions can enhance student engagement, reduce failure rates, and ultimately improve the quality of education.

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